

Heat as a witness: Certifying properties of quantum states via heat transfer

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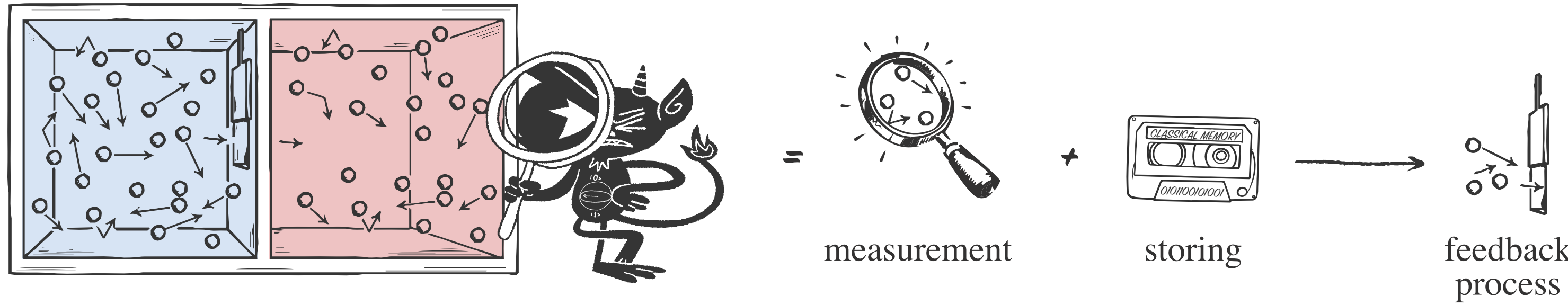
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Can we infer the presence of **quantum features** via **thermodynamic quantities**?

I. Rough formulation of the problem

Maxwell's demon as a thermodynamic protocol:



M. B. Plenio and V. Vitelli, *Contemp. Phys* 42, 25 (2001)

What if the **memory** is a **quantum system**?

Quantum memory: "unspeakable" information + can become entangled with the system!

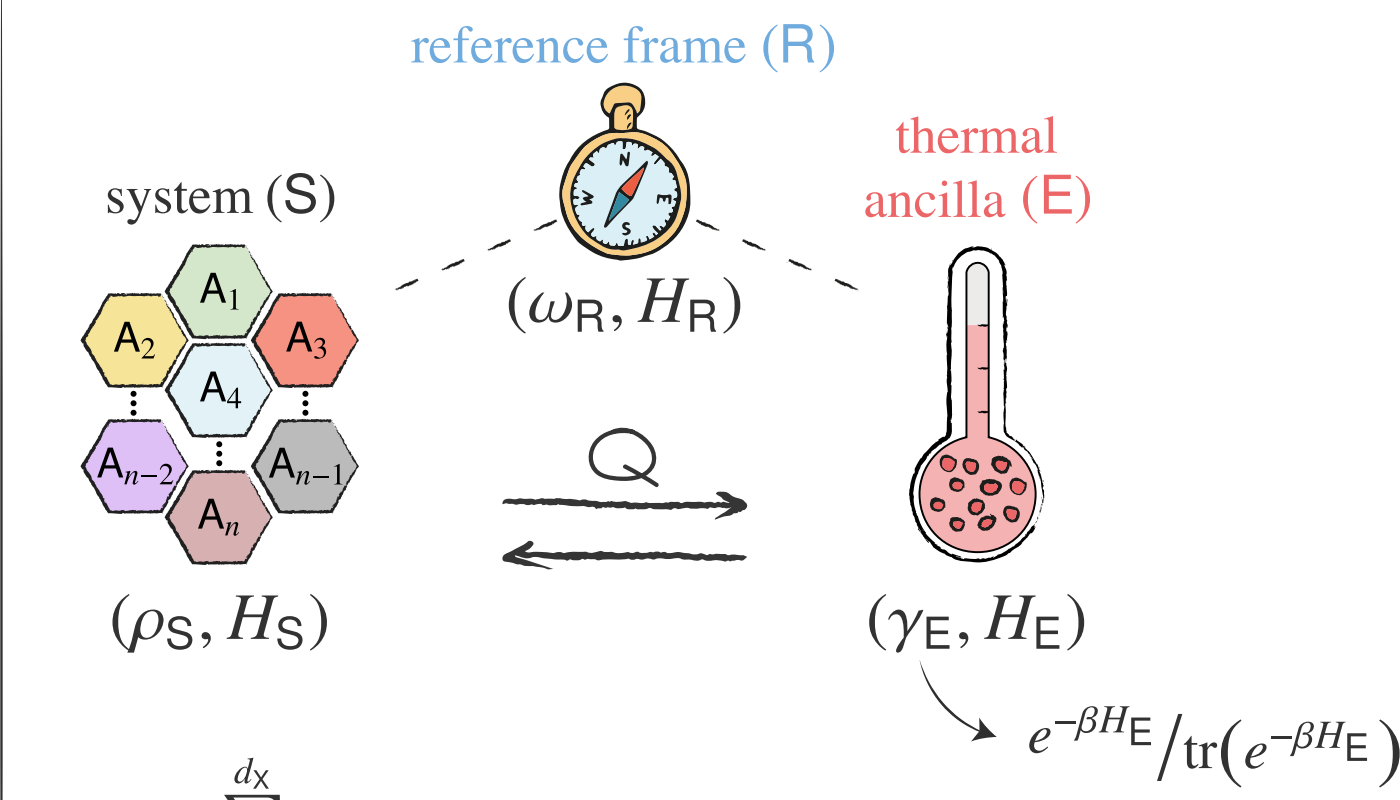
reference frame

I. Marvian et al, *Phys. Rev. A* 94, 052324 (2016)

How can we **model** such a scenario?

II. Framework

Thermodynamic setup:



$$H_X = \sum_{i=1}^{d_X} \epsilon_i^X |i\rangle\langle i|_X \text{ with } X \in \{S, R, E\}$$

$$E_S = \text{tr}(H_S \rho_S)$$

P. Lipka-Bartosik et al, *Phys. Rev. Lett* 130, 040401 (2023)

P. Lipka-Bartosik et al, *Phys. Rev. Lett* 132, 140402 (2024)

A. de Oliveira Junior et al, *Phys. Rev. Research* 6, 023127 (2024)

D. Janzing et al, *Int. J. Theor. Phys.* 39, 2717 (2000)

i. SRE evolves under an energy-preserving unitary process:

$$\eta_{\text{SRE}} = U(\rho_S \otimes \omega_R \otimes \gamma_E)U^\dagger, \text{ where } [U, H_S + H_R + H_E] = 0.$$

ii. We demand that:

$$\text{tr}_{\text{SE}}[U(\rho_S \otimes \omega_R \otimes \gamma_E)U^\dagger] = \omega_R.$$

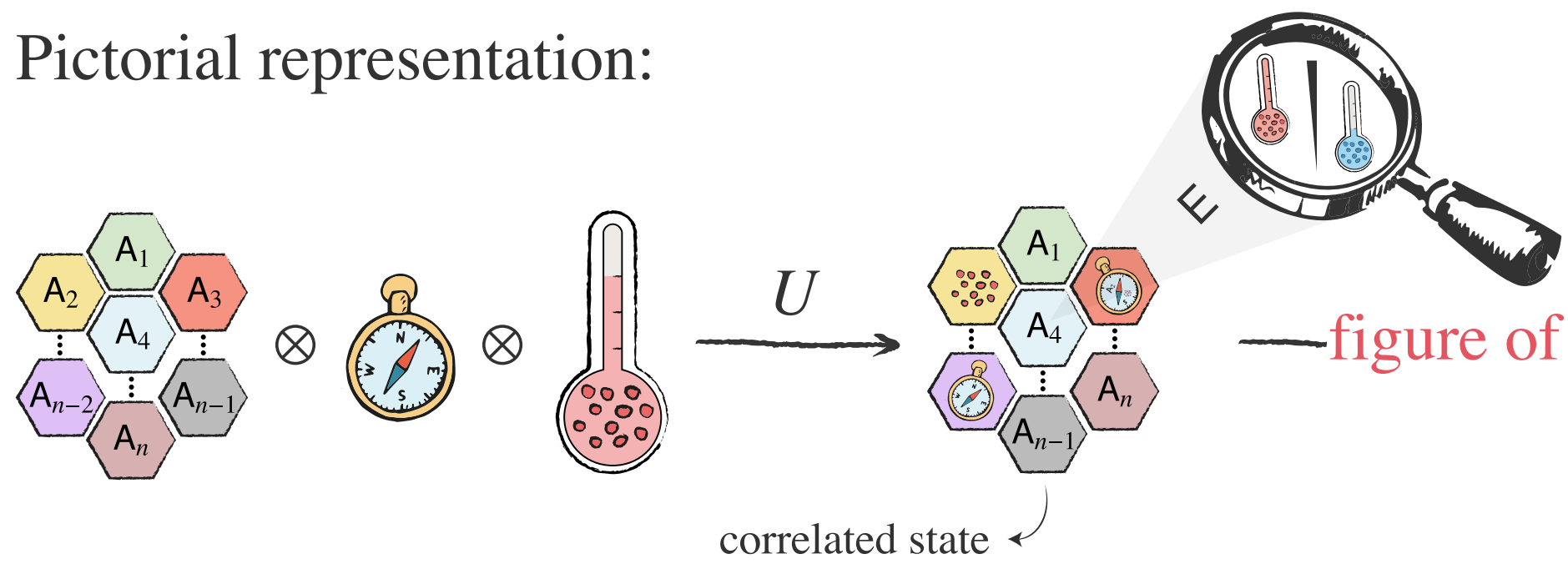
iii. Probe the heat exchange between SE:

$$Q_{c/h}(\rho) := \text{tr}[H_E(\eta_E - \gamma_E)].$$

What is the **optimal** heat flow?

II. Optimal heat flow with quantum feedback

Pictorial representation:

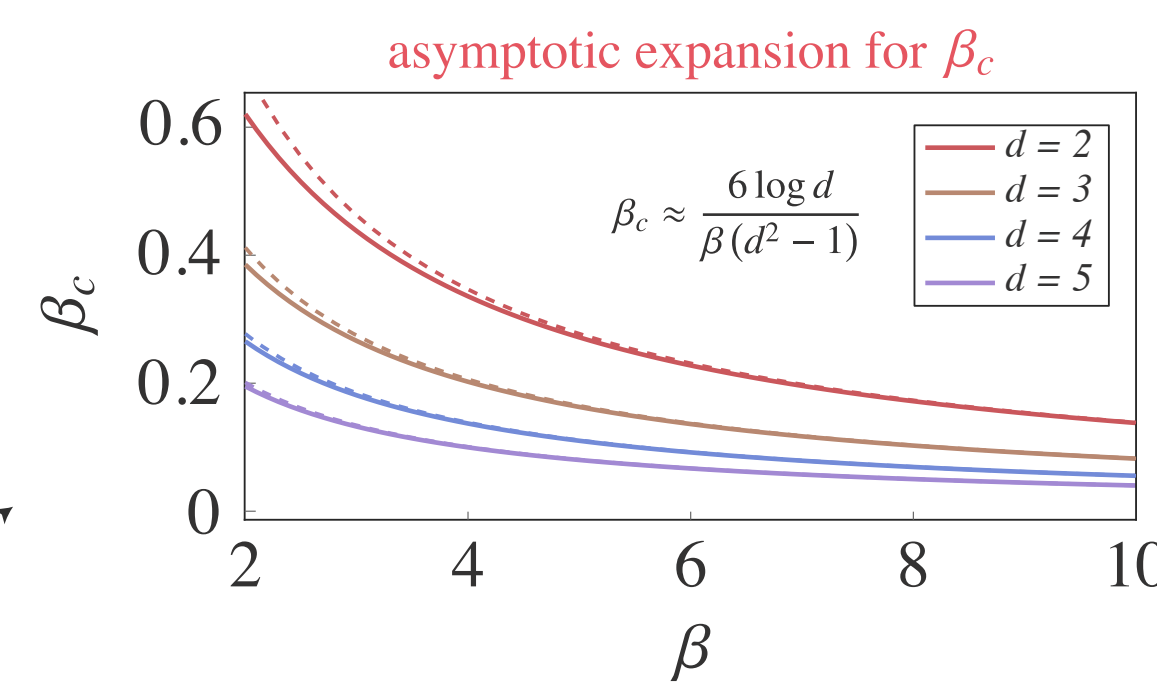


$$Q_{c/h}(\rho_S) := \min_{H_E, H_R, U, \omega_R} \max_{\eta_S} \text{tr}[H_E(\eta_E - \gamma_E)], \text{ s.t. } [U, H_S + H_R + H_E] = 0, \eta_R = \omega_R.$$

Urgh, very nasty!

$$Q_{c/h}(\rho_S) := \min_{\eta_S} \max_{\eta_S} \text{tr}[H_S(\rho_S - \eta_S)], \text{ s.t. } F_\beta(\rho_S) \geq F_\beta(\eta_S).$$

that's cool!



Simple solution:

$$Q_c(\rho) = \beta^{-1} \{S(\rho) - S[\gamma(\beta_c)]\},$$

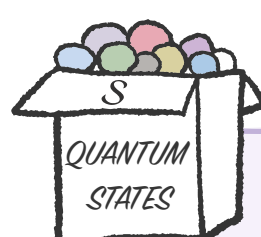
$$Q_h(\rho) = \min \{ \beta^{-1} [S(\rho) - S(\gamma(\beta_h))], \text{tr}[H_S \rho] \}.$$

★ $\gamma(\beta_{c/h})$: thermal state at inverse temperature $\beta_{c/h}$.

★ The inverse temperatures $\beta_{c/h}$ are zeros of the function: $f(x, y) := y - \left(1 - \frac{x}{\beta}\right) \text{tr}[H_S \gamma(x)] + \frac{1}{\beta} \log Z(x)$.

III. Certifying properties of quantum states via heat exchange

Can **heat flow** mediated by the reference frame reveal **properties** of the system?



i. Let S be a fixed set of quantum states:

$$Q_c(S) := \min_{\rho \in S} Q_c(\rho), \quad Q_h(S) := \max_{\rho \in S} Q_h(\rho).$$

ii. Any quantum state in S satisfies:

$$Q_c(S) \leq Q(\rho) \leq Q_h(S).$$

iii. If we know $F_\beta^*(S)$, such that $\max_{\rho \in S} F_\beta(\rho) \leq F_\beta^*(S)$, then

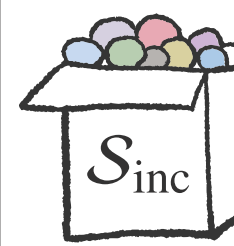
$$Q_c(S) \geq \beta^{-1} [S(\rho) - S[\gamma(\beta_c^*)]] =: Q_c^*(S).$$

witness for the set S

★ Two central examples:



$$F_\beta(\rho_{AB}) = F_\beta(\rho_A) + F_\beta(\rho_B) + \frac{1}{\beta} I(A : B)_\rho.$$



$$F_\beta(\rho_S) = F_\beta(\tilde{\rho}_S) + D(\rho_S \| \tilde{\rho}_S).$$

★ In both cases, we can find $F_\beta^*(S_{\text{sep/inc}})$ and, therefore, $Q_{c/h}^*(S_{\text{sep/inc}})$ which acts as a witness for $S_{\text{sep/inc}}$.

Informational and thermodynamic quantities

- ★ $S(\rho) := -\text{tr}(\rho \log \rho)$ von Neumann entropy
- ★ $S(A|B)_\rho := S(\rho_{AB}) - S(\rho_B)$ conditional entropy
- ★ $D(\rho \| \sigma) := \text{tr}[\rho(\log \rho - \log \sigma)]$ relative entropy
- ★ $I(A : B)_\rho := S(\rho_B) - S(A|B)_\rho$ mutual information
- ★ $\tilde{\rho} := \sum_{i=1}^n \langle i | \rho | i \rangle |i\rangle\langle i|$ dephased state with respect to $|i\rangle$
- ★ $h(x) := -x \log x - (1-x) \log(1-x)$ binary entropy
- ★ $Z(x) := \text{tr}(e^{-xH})$ partition function
- ★ $F_\beta(\rho) := \text{tr}[\rho H] - \beta^{-1} S(\rho)$ nonequilibrium free-energy

N. Cerf et al, *Springer* (1997)

N. Cerf et al, *Phys. Rev. A* 60, 893 (1999)

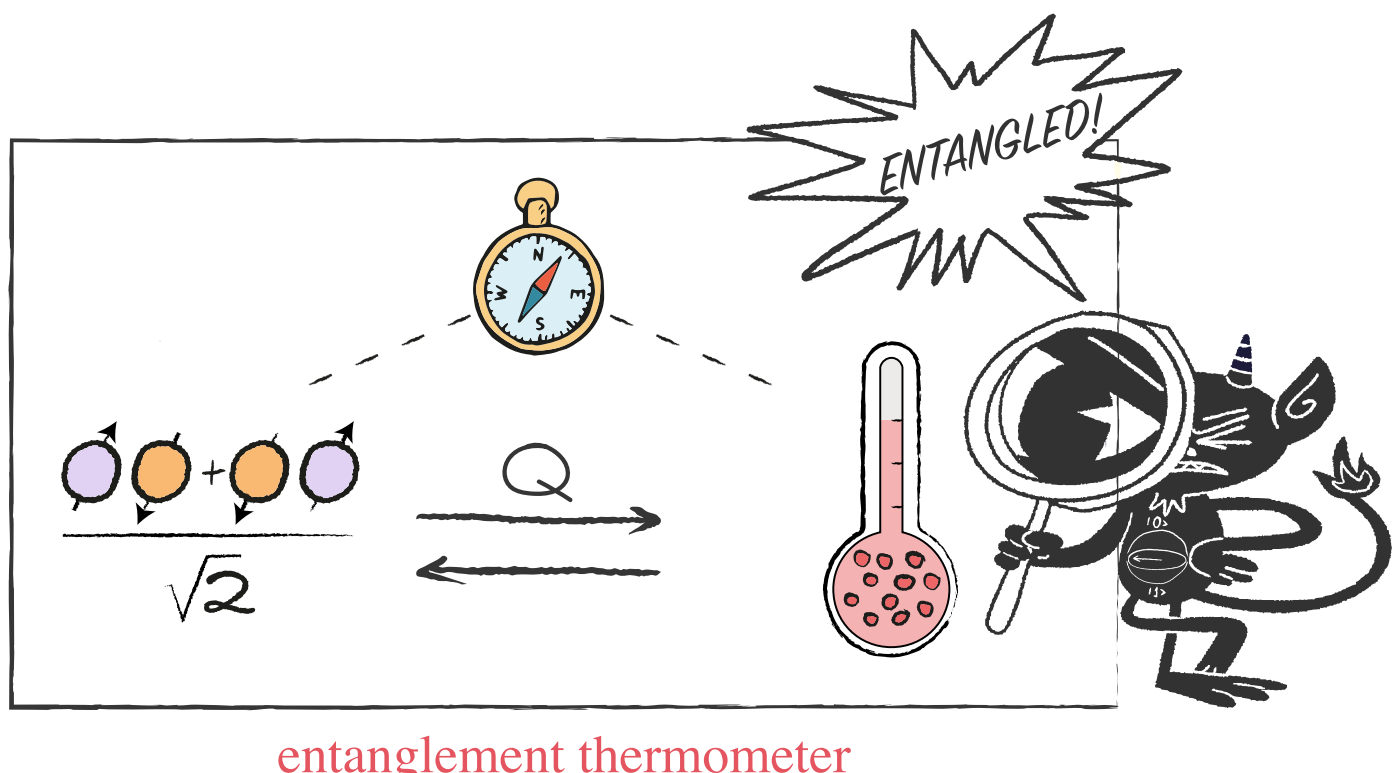
M. Lostaglio et al, *Nat. Commun* 6, 6383 (2015)

D. Janzing, *J. Stat. Phys.* 125, 761 (2006)

Examples

Entanglement witness

★ Bipartite system $S = AB$ and $S = S_{\text{sep}}$: $F_\beta^*(S_{\text{sep}}) = E(\rho_A) + E(\rho_B) - \beta^{-1} \max\{S(\rho_A), S(\rho_B)\}$.



entanglement thermometer

★ Classical heat exchange:

$$Q_{c/h}^*(S_{\text{sep}}) = \beta^{-1} [\max\{S(\rho_A), S(\rho_B)\} - S[\gamma_{AB}(\beta_{c/h}^*)]].$$

★ Heat-based entanglement witness:

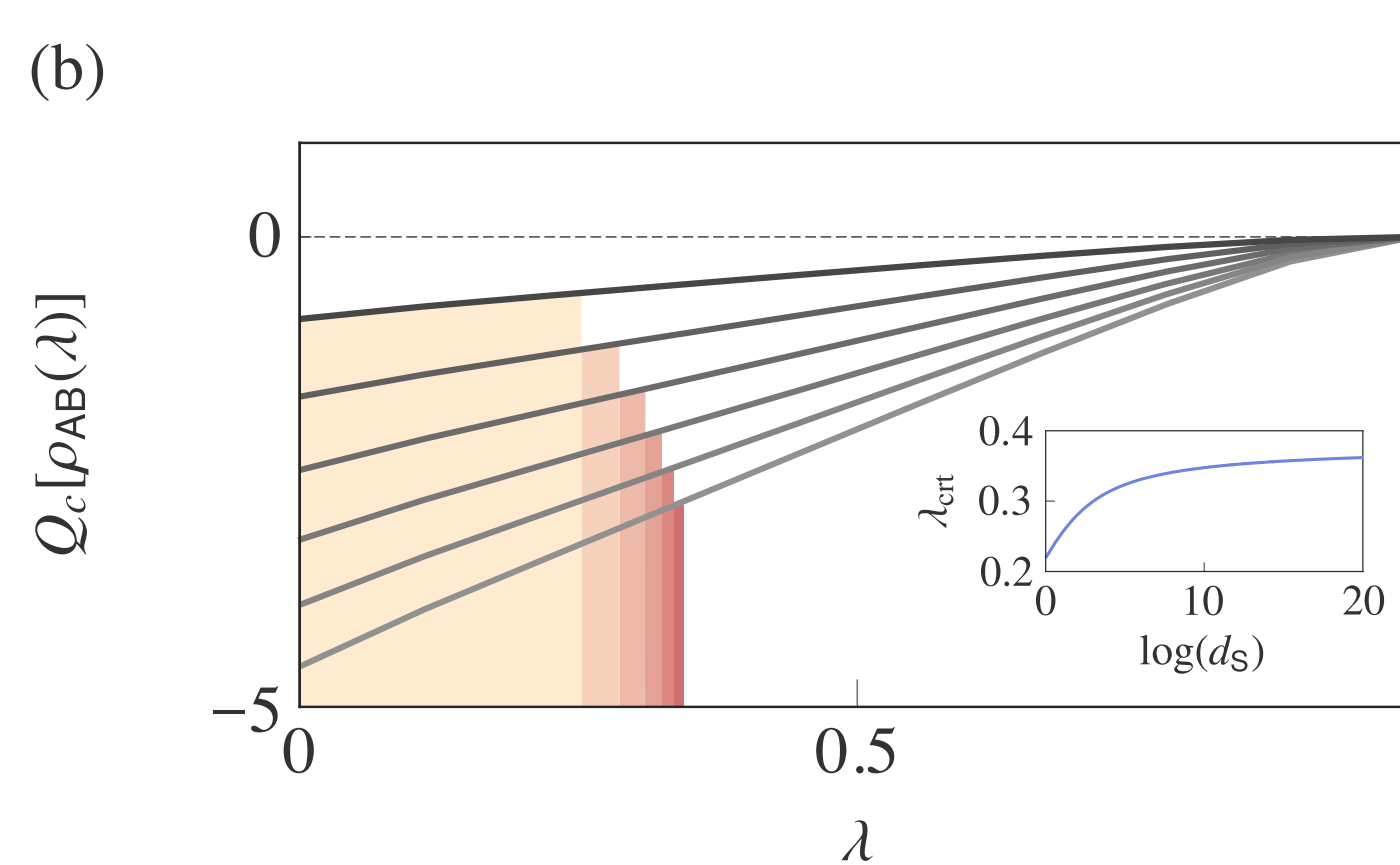
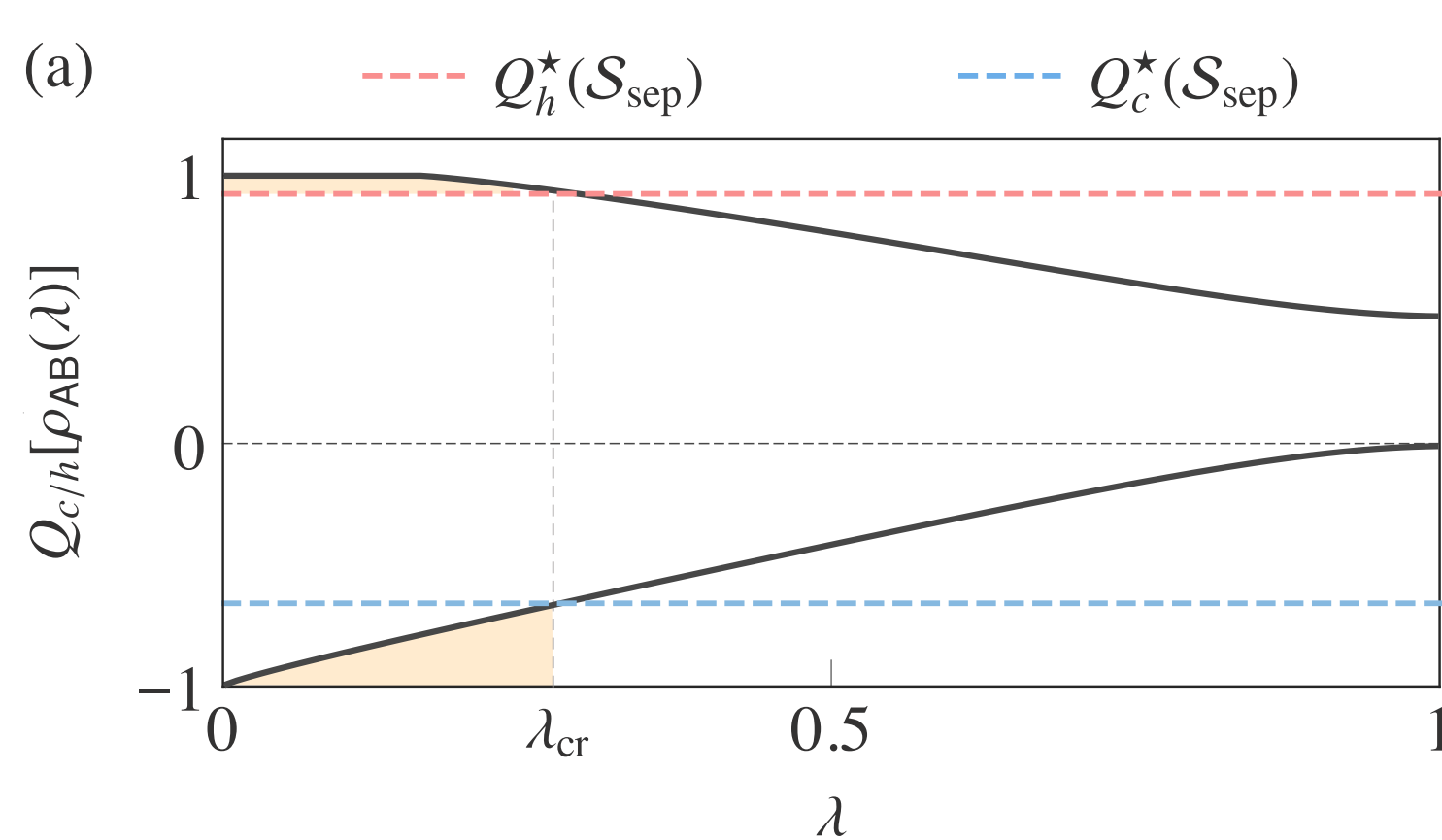
$$Q_c^*(S_{\text{sep}}) \leq Q(\rho_{AB}) \leq Q_h^*(S_{\text{sep}}).$$

★ Can detect **any** bipartite pure entangled state!

How about for **mixed states**?

★ Isotropic state: $\rho_{AB}(\lambda) := (1-\lambda) |\psi^+\rangle\langle\psi^+|_{AB} + \lambda \frac{\mathbb{1}_{AB}}{d^2}$.

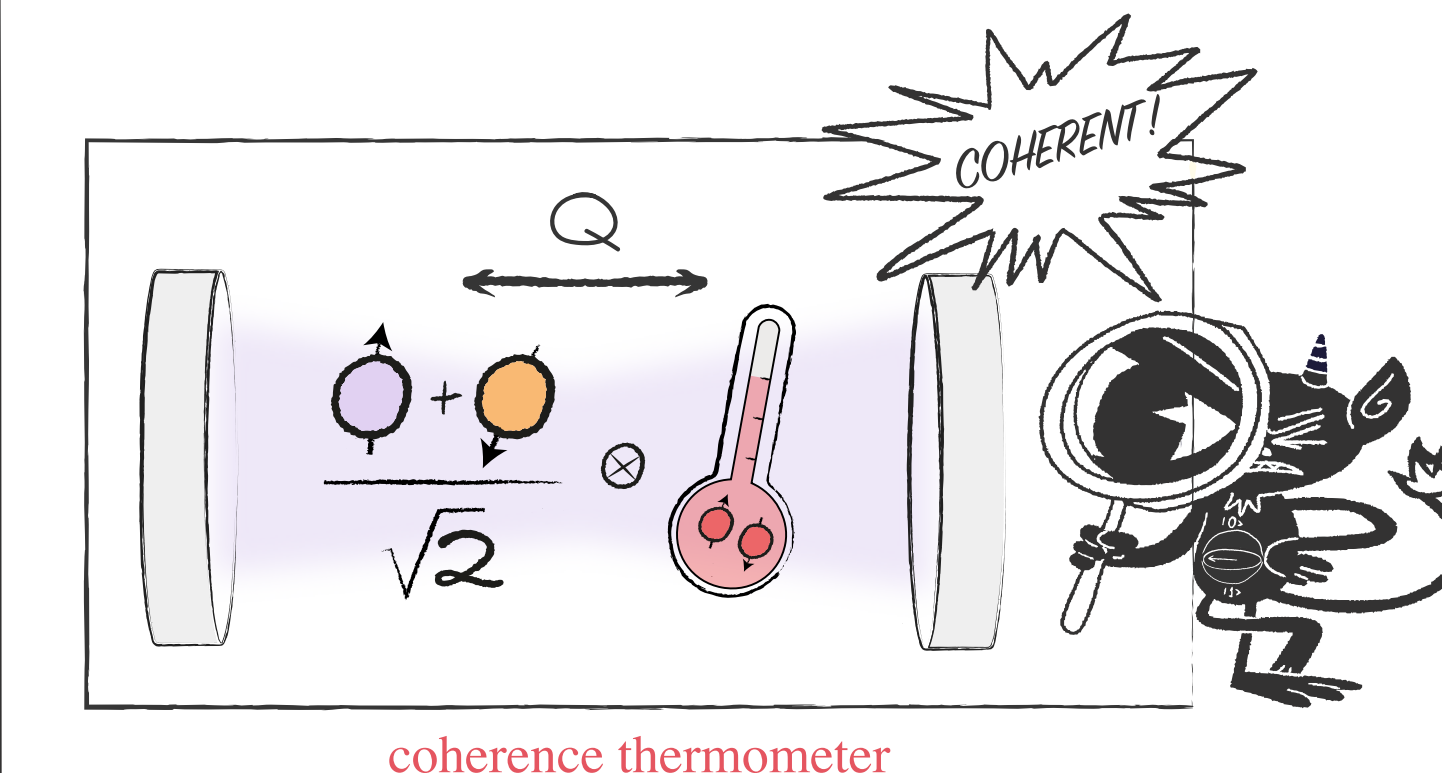
max entangled state $|\psi^+\rangle_{AB} := d^{-1/2} \sum_{i=1}^d |i\rangle_A |i\rangle_B$.



★ Results also hold for **multipartite system**!

Certification of quantum coherence

★ Coherent state and S_{inc} : $F_\beta^*(S_{\text{inc}}) := E_S - \beta^{-1} h(E_S/\epsilon_d)$



coherence thermometer

★ Classical heat exchange:

$$Q_{c/h}^*(S_{\text{inc}}) = \beta^{-1} [S(\rho_S) - S[\gamma_S(\beta_{c/h}^*)]].$$

★ Heat-based coherence witness:

$$Q_c^*(S_{\text{inc}}) \leq Q(\rho_S) \leq Q_h^*(S_{\text{inc}}).$$

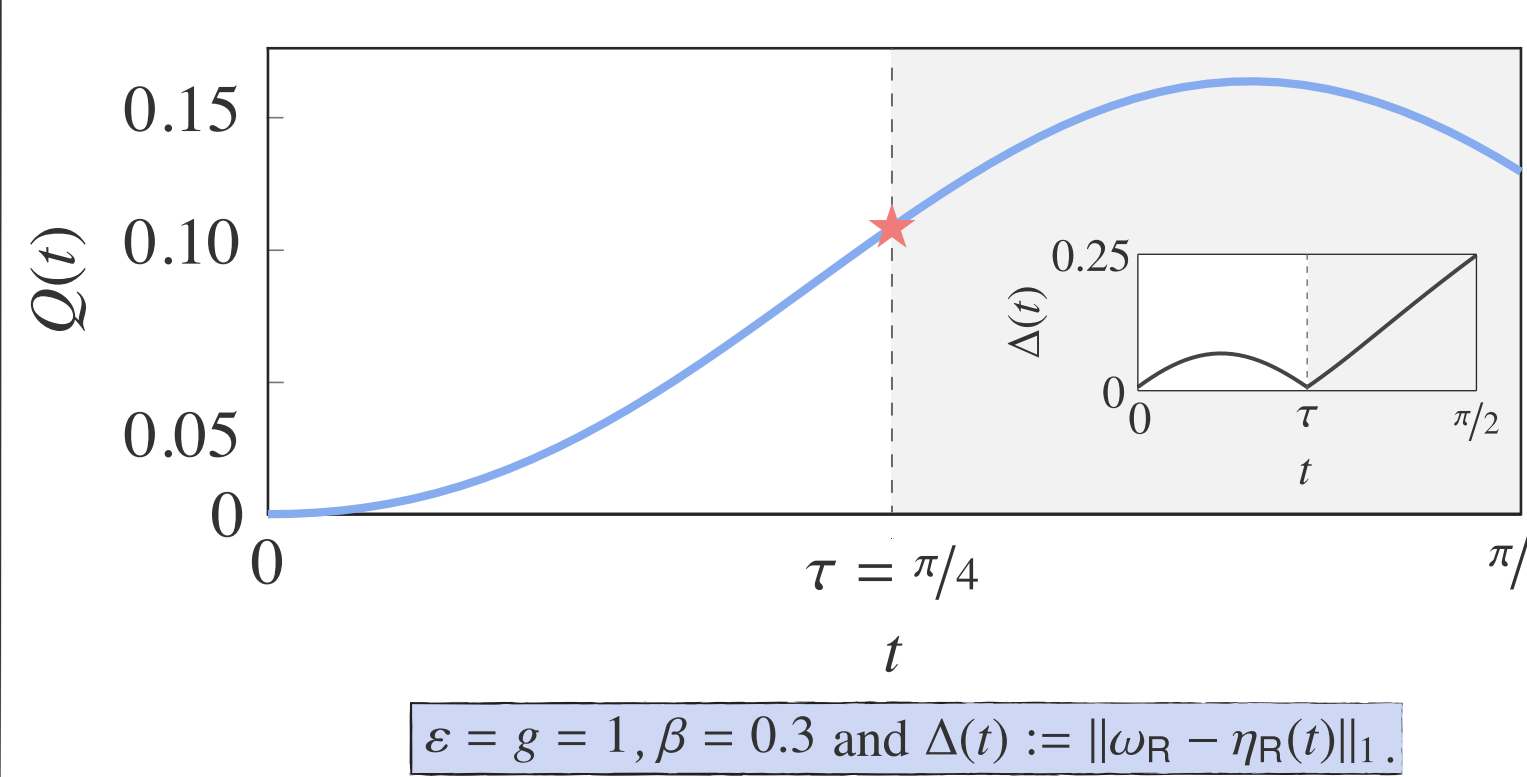
★ Two spins coupled to an electromagnetic field

$$H_{\text{SRE}} = \varepsilon (\sigma_S^\dagger \sigma_S + \sigma_E^\dagger \sigma_E + a_R^\dagger a_R) + g (a_R \sigma_S^\dagger + a_R \sigma_E^\dagger + \text{c.c.}).$$

★ System prepared at

$$|\psi\rangle_S := (|0\rangle_S + e^{-\beta H_S/2} |1\rangle_S) / \sqrt{1 + e^{-\beta H_S/2}}.$$

★ $Q_{c/h}^*(S_{\text{inc}}) = 0$



★ Impossible with classical memory lacking energy coherence.

★ Access to correlations locked in the degenerate energy-subspace!

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